# Importance Sampling in Off-Policy Learning

## 1. What is Importance Sampling?

Importance sampling is a statistical technique used to estimate the expected value of a random variable under one probability distribution using samples generated from a different distribution. In reinforcement learning, it allows us to estimate the value of a *target policy* while episodes are generated by a different *behavior policy.* The key idea is to reweight observed returns so that, in expectation, they correspond to the target policy rather than the behavior policy.

## 2. Why Importance Sampling is Required in Off-Policy Methods

In off-policy learning, the agent does not follow the policy whose value it wishes to evaluate. Instead, data is generated using a more exploratory behavior policy. Directly averaging returns from such data yields estimates corresponding to the behavior policy, not the target policy. Importance sampling corrects this mismatch by weighting each return according to how likely the observed trajectory is under the target policy relative to the behavior policy. This correction enables learning about optimal or deterministic policies while behaving stochastically.

## 3. Off-Policy Prediction Problem

The off-policy prediction problem aims to estimate the state-value function $v_\pi(s)$ or action-value function $q_\pi(s, a)$ for a fixed target policy $\pi$, given episodes generated by a different fixed behavior policy $b$. A key requirement is the *coverage condition*: if $\pi(a|s) > 0$, then $b(a|s) > 0$. This ensures that all actions required by the target policy are sampled, at least occasionally, by the behavior policy.

## 4. Importance Sampling Ratio

Consider a trajectory starting at time $t$ and terminating at time $T$. The importance sampling ratio for this trajectory is defined as

$$\rho_{t:T-1} = \prod_{k=t}^{T-1} \frac{\pi(A_k|S_k)}{b(A_k|S_k)}.$$

This ratio measures how much more (or less) likely the observed sequence of actions is under the target policy compared to the behavior policy. The transition probabilities cancel out, making the ratio depend only on the policies.

Using this ratio, an unbiased estimator of $v_\pi(s)$ is obtained as

$$\mathbb{E}[\rho_{t:T-1} G_t \mid S_t = s] = v_\pi(s),$$

where $G_t$ is the return following time $t$.

# 5. Example 1: Estimating $q_\pi(s, a)$ from a Single Episode

Consider an episodic MDP with two states $\{s_0, s_1\}$ and two actions $\{a_1, a_2\}$. Let the target policy $\pi$ be deterministic, choosing $a_1$ in both states. Let the behavior policy $b$ choose each action with probability 0.5.

Suppose the following episode of length 6 is generated under $b$:

$$(s_0, a_1, 1, s_1),\ (s_1, a_2, 0, s_0),\ (s_0, a_1, 2, s_1),\ (s_1, a_1, 1, s_1),\ (s_1, a_2, -1, s_0),\ (s_0, a_1, 3, \text{terminal})$$

Assuming $\gamma = 1$, the return following the first occurrence of $(s_0, a_1)$ is

$$G = 1 + 0 + 2 + 1 - 1 + 3 = 6.$$

The importance sampling ratio for this trajectory segment is

$$\rho = \prod \frac{\pi(a_k|s_k)}{b(a_k|s_k)}.$$

Since $\pi(a_2|s_1) = 0$, the ratio becomes zero after the first $a_2$ is taken, and hence this episode contributes zero weight to the estimate of $q_\pi(s_0, a_1)$.

## With Discounting

If $\gamma = 0.9$, the return becomes

$$G = 1 + 0.9(0) + 0.9^2(2) + 0.9^3(1) + 0.9^4(-1) + 0.9^5(3).$$

The importance sampling ratio remains unchanged; discounting affects only the return, not the ratio.

# 6. Example 2: Ordinary vs Weighted Importance Sampling

Suppose three episodes generate returns $G_1, G_2, G_3$ for state $s$, with corresponding ratios $\rho_1, \rho_2, \rho_3$.

**Ordinary importance sampling:**

$$V(s) = \frac{1}{3}\sum_{i=1}^{3} \rho_i G_i$$

This estimator is unbiased but may have very high variance.

**Weighted importance sampling:**

$$V(s) = \frac{\sum_{i=1}^{3} \rho_i G_i}{\sum_{i=1}^{3} \rho_i}$$

This estimator is biased for finite samples, but the bias vanishes asymptotically and the variance is significantly lower.

## 7. Exercises

**Exercise 1.** Consider an off-policy setting with a deterministic target policy and an $\epsilon$-greedy behavior policy. Explain why the coverage condition is satisfied and compute the importance sampling ratio for a 4-step episode.

**Exercise 2.** Given two episodes with returns $G_1 = 5$, $G_2 = 2$ and importance sampling ratios $\rho_1 = 4$, $\rho_2 = 0.5$, compute the value estimate using (a) ordinary importance sampling and (b) weighted importance sampling. Comment on the difference.

# Ordinary vs. Weighted Importance Sampling

## 1. Background and Setting

In off-policy Monte Carlo prediction, we wish to estimate the value of a target policy $\pi$ using episodes generated by a different behavior policy $b$. Because the observed returns are distributed according to $b$, they must be corrected to reflect expectations under $\pi$. Importance sampling provides this correction by weighting each return by the relative likelihood of the observed trajectory under the two policies.

Throughout this document, we assume the coverage condition holds:

$$\pi(a|s) > 0 \implies b(a|s) > 0,$$

ensuring that importance sampling ratios are always well-defined and nonzero.

## 2. Importance Sampling Ratio

For a trajectory segment starting at time $t$ and terminating at time $T$, the importance sampling ratio is defined as

$$\rho_{t:T-1} = \prod_{k=t}^{T-1} \frac{\pi(A_k \mid S_k)}{b(A_k \mid S_k)}.$$

This ratio measures how much more (or less) likely the observed action sequence is under the target policy compared to the behavior policy. Note that environment transition probabilities cancel out and do not appear in the ratio.

## 3. Ordinary Importance Sampling

Ordinary importance sampling estimates the value function by averaging the importance-weighted returns:

$$V(s) = \frac{1}{N} \sum_{i=1}^{N} \rho_i G_i,$$

where $G_i$ is the return observed after a visit to state $s$ and $\rho_i$ is the corresponding importance sampling ratio.

### First-Visit Ordinary Importance Sampling

In the first-visit setting, only the return following the first occurrence of state $s$ in each episode is used. This estimator is **unbiased**, meaning

$$\mathbb{E}[\rho G] = v_\pi(s),$$

but it can suffer from extremely high variance when ratios are large.

### Every-Visit Ordinary Importance Sampling

In the every-visit setting, all occurrences of state $s$ in each episode contribute returns. Although simpler to implement, the estimator becomes biased for finite samples, though the bias vanishes asymptotically.

## 4. Weighted Importance Sampling

Weighted importance sampling normalizes the importance-weighted returns:

$$V(s) = \frac{\sum_{i=1}^{N} \rho_i G_i}{\sum_{i=1}^{N} \rho_i}.$$

This normalization limits the influence of any single return and greatly reduces variance.

### First-Visit Weighted Importance Sampling

For first-visit weighted importance sampling, only first visits are included in the numerator and denominator. The estimator is **biased** for finite samples, but the bias converges to zero as the number of episodes increases.

### Every-Visit Weighted Importance Sampling

Every-visit weighted importance sampling uses all visits to the state and is widely preferred in practice due to its low variance and ease of implementation, especially in large or continuing tasks.

## 5. Example 1: Single Episode, First-Visit Estimation

Consider an episodic MDP with states $\{s_0, s_1\}$ and actions $\{a_1, a_2\}$. Let the target policy $\pi$ and behavior policy $b$ be defined as follows:

$$\pi(a_1|s) = 0.7, \quad \pi(a_2|s) = 0.3,$$

$$b(a_1|s) = 0.5, \quad b(a_2|s) = 0.5 \quad \text{for all } s.$$

Suppose the first visit to $s_0$ occurs at time $t$ and the episode continues for three steps with actions

$$(a_1, a_2, a_1)$$

and rewards

$$(2, 0, 1),$$

with $\gamma = 1$.

The return is

$$G = 2 + 0 + 1 = 3.$$

The importance sampling ratio is

$$\rho = \frac{0.7}{0.5} \cdot \frac{0.3}{0.5} \cdot \frac{0.7}{0.5} = 1.176.$$

**Ordinary IS estimate:**

$$V(s_0) = 1.176 \times 3 = 3.528.$$

**Weighted IS estimate (single sample):**

$$V(s_0) = \frac{1.176 \times 3}{1.176} = 3.$$

# 6. Example 2: Multiple Episodes, Every-Visit Estimation

Suppose two episodes produce returns for state $s_1$ as follows:

$$(G_1, \rho_1) = (4, 1.4), \quad (G_2, \rho_2) = (2, 0.8).$$

**Every-visit ordinary importance sampling:**

$$V(s_1) = \frac{1}{2}(1.4 \cdot 4 + 0.8 \cdot 2) = \frac{1}{2}(5.6 + 1.6) = 3.6.$$

**Every-visit weighted importance sampling:**

$$V(s_1) = \frac{1.4 \cdot 4 + 0.8 \cdot 2}{1.4 + 0.8} = \frac{7.2}{2.2} \approx 3.27.$$

This example illustrates how weighted importance sampling moderates the effect of large ratios and produces a more stable estimate.

# 7. Summary Comparison

- Ordinary importance sampling is unbiased but may have unbounded variance.
- Weighted importance sampling is biased for finite samples but has much lower variance.
- First-visit methods are theoretically cleaner but harder to implement.
- Every-visit methods are simpler and commonly used in practice.

# Bias and Variance in Ordinary and Weighted Importance Sampling

## 1. Context and Objective

In off-policy Monte Carlo prediction, we estimate the value of a target policy $\pi$ using episodes generated by a different behavior policy $b$. Because the observed returns are drawn from the wrong distribution, importance sampling is used to correct the mismatch. Two commonly used estimators are *ordinary importance sampling* and *weighted importance sampling*. Their key difference lies in their bias and variance properties.

## 2. Importance Sampling Estimators

Let $G_i$ denote the return following a visit to state $s$ (or stateaction pair $(s, a)$) in episode $i$, and let $\rho_i$ be the corresponding importance sampling ratio.

### Ordinary Importance Sampling

The ordinary importance sampling estimator is defined as

$$V_{\text{OIS}}(s) = \frac{1}{N} \sum_{i=1}^{N} \rho_i G_i.$$

This estimator directly averages the importance-weighted returns.

### Weighted Importance Sampling

The weighted importance sampling estimator is defined as

$$V_{\text{WIS}}(s) = \frac{\sum_{i=1}^{N} \rho_i G_i}{\sum_{i=1}^{N} \rho_i},$$

with the convention that the estimate is zero if the denominator is zero.

## 3. Bias Properties

### Ordinary Importance Sampling

Ordinary importance sampling is **unbiased**. Formally,

$$\mathbb{E}[\rho_i G_i \mid S = s] = v_\pi(s),$$

and therefore

$$\mathbb{E}[V_{\text{OIS}}(s)] = v_\pi(s).$$

This property holds for the first-visit estimator and does not depend on the number of samples.

### Weighted Importance Sampling

Weighted importance sampling is **biased** for finite samples because the expectation of a ratio of random variables is not equal to the ratio of expectations:

$$\mathbb{E}\left[\frac{\sum \rho_i G_i}{\sum \rho_i}\right] \neq v_\pi(s).$$

However, the bias converges asymptotically to zero as $N \to \infty$, making the estimator consistent.

### Every-Visit Case

In the every-visit setting, both ordinary and weighted importance sampling estimators are biased for finite samples. As the number of visits increases, the bias decreases and converges to zero asymptotically.

## 4. Variance Properties

### Ordinary Importance Sampling

The variance of ordinary importance sampling can be **unbounded**. This occurs because the importance sampling ratios $\rho_i$ are products of policy probability ratios and can become arbitrarily large. Even if the returns $G_i$ are bounded, large ratios can cause extreme variance:

$$\text{Var}(\rho_i G_i) \to \infty \quad \text{if } \text{Var}(\rho_i) \to \infty.$$

### Weighted Importance Sampling

In weighted importance sampling, the effective weight on any single return is

$$\frac{\rho_i}{\sum_{j=1}^{N} \rho_j} \leq 1.$$

As a result, assuming bounded returns, the variance of the weighted estimator converges to zero as the number of samples increases, even if the variance of the ratios themselves is infinite (Precup, Sutton, and Dasgupta, 2001). This makes weighted importance sampling far more stable in practice.

# 5. Numerical Example

Suppose two episodes produce returns for state $s$ as follows:

$$(G_1, \rho_1) = (10, 8), \quad (G_2, \rho_2) = (2, 1).$$

## Ordinary Importance Sampling

$$V_{\text{OIS}}(s) = \frac{1}{2}(8 \cdot 10 + 1 \cdot 2) = \frac{82}{2} = 41.$$

This estimate is strongly influenced by the large ratio $\rho_1$.

## Weighted Importance Sampling

$$V_{\text{WIS}}(s) = \frac{8 \cdot 10 + 1 \cdot 2}{8 + 1} = \frac{82}{9} \approx 9.11.$$

Here, the normalization limits the influence of the large ratio, producing a more stable estimate.

# 6. Practical Implications

- Ordinary importance sampling is unbiased but may have extremely high or unbounded variance.
- Weighted importance sampling is biased for finite samples but has much lower variance.
- In practice, weighted importance sampling is strongly preferred.
- Ordinary importance sampling remains useful because it is easier to extend to function approximation and other approximate methods.

# 7. Summary

The fundamental trade-off between ordinary and weighted importance sampling is a trade-off between unbiasedness and variance. Ordinary importance sampling preserves correctness in expectation but may be unstable, while weighted importance sampling sacrifices unbiasedness for dramatically improved variance properties, making it the method of choice in most practical off-policy Monte Carlo algorithms.

# Monte Carlo Prediction: First-Visit vs Every-Visit (Single-State MDP)

## Problem Statement

Consider an episodic MDP with a single nonterminal state $s$ and a single action $a$. After taking $a$ in $s$, the process returns to $s$ with probability $p$ and terminates with probability $1-p$. Every transition gives reward $+1$, and the discount factor is $\gamma = 1$.

You observe exactly one episode that lasts 10 steps (i.e., 10 transitions), so the total return from the beginning of the episode is $G_0 = 10$.

**Question:** Using this single episode, what are the **first-visit** and **every-visit** Monte Carlo estimators of $V(s)$?

## Simple Explanation (What is happening?)

Because there is only one nonterminal state $s$, the episode looks like:

$$s \to s \to s \to \cdots \to s \to \text{terminal},$$

and since each step gives reward $+1$ and $\gamma = 1$, the return from any time step is simply the number of remaining steps until termination.

So:

$$G_0 = 10,\ G_1 = 9,\ G_2 = 8,\ \ldots,\ G_9 = 1.$$

## How to Approach Answering (Step-by-step)

1. **Identify visits to the state $s$:** here, $s$ is visited at every time step $t = 0, 1, \ldots, 9$.
2. **Compute returns:** with $\gamma = 1$ and reward $+1$ each step, $G_t$ equals remaining steps.
3. **Apply estimator definition:**
   - First-visit MC uses only the return after the *first* visit to $s$ in the episode.
   - Every-visit MC averages returns after *all* visits to $s$ in the episode.

# Solution

## First-Visit MC Estimator

The first visit to $s$ in the episode is at $t = 0$. Therefore the first-visit estimator from this single episode is:

$$V_{\text{FV}}(s) = G_0 = 10.$$

## Every-Visit MC Estimator

Every-visit MC uses returns following *all* visits to $s$ in the episode, i.e., $t = 0$ to 9:

$$G_0 = 10,\ G_1 = 9,\ \ldots,\ G_9 = 1.$$

So the every-visit estimator is the average:

$$V_{\text{EV}}(s) = \frac{1}{10}\sum_{t=0}^{9} G_t = \frac{1}{10}(10 + 9 + 8 + \cdots + 1).$$

We know:

$$10 + 9 + \cdots + 1 = \frac{10 \cdot 11}{2} = 55.$$

Hence,

$$V_{\text{EV}}(s) = \frac{55}{10} = 5.5.$$

# Final Answer

$$\boxed{V_{\text{FV}}(s) = 10} \qquad \boxed{V_{\text{EV}}(s) = 5.5}$$

**Refer to Sutton & Barto Prediction Problem for Off-Policy Monte-Carlo - Algorithm**

## Off-policy MC Prediction (Q-evaluation) with Importance Sampling

**Setup.** Two states $\mathcal{S} = \{s_1, s_2\}$, two actions $\mathcal{A} = \{a_1, a_2\}$. We evaluate target policy $\pi$ using episodes generated by behavior policy $b$.

$$\pi(a_1 \mid s_1) = 1,\ \pi(a_2 \mid s_2) = 1,\ \text{else } 0 \qquad b(a_1 \mid s) = 0.6,\ b(a_2 \mid s) = 0.4\ \ (\forall s \in \{s_1, s_2\})$$

Coverage holds since $b(a \mid s) > 0$ where $\pi(a \mid s) > 0$.
Initialize for all $(s, a)$: $Q(s, a) = 0$, $C(s, a) = 0$. No discounting (as in the given algorithm).

**Three episodes (generated by $b$).** Each episode is $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$ and then terminates.

$$\begin{aligned}
&\text{Episode 1: } (s_1, a_2, r_1 = 1)\ \to\ (s_2, a_2, r_2 = 2)\ \to\ \text{terminal}\\
&\text{Episode 2: } (s_1, a_1, r_1 = 0)\ \to\ (s_1, a_1, r_2 = 3)\ \to\ \text{terminal}\\
&\text{Episode 3: } (s_2, a_2, r_1 = -1)\ \to\ (s_1, a_1, r_2 = 2)\ \to\ \text{terminal}
\end{aligned}$$

We apply the off-policy MC algorithm *backwards* in each episode with:

$$\begin{aligned}
&G \leftarrow G + R_{t+1}, \quad C(S_t, A_t) \leftarrow C(S_t, A_t) + W,\\
&Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)}\big(G - Q(S_t, A_t)\big), \quad W \leftarrow W \cdot \frac{\pi(A_t \mid S_t)}{b(A_t \mid S_t)}
\end{aligned}$$

### Walk-through (backward in time)

Below, $w_t = \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$, and we show $W$ before multiplying by $w_t$.

**Episode 1.**

| | |
|---|---|
| Step $t = 1 : (s_2, a_2),\ R_2 = 2$ | $G \leftarrow 0 + 2 = 2,\ \ C(s_2, a_2) \leftarrow 0 + 1 = 1$<br>$Q(s_2, a_2) \leftarrow 0 + \frac{1}{1}(2 - 0) = 2$<br>$w_1 = \frac{1}{0.4} = \frac{5}{2},\ \ W:\ 1 \mapsto \frac{5}{2}$ |
| Step $t = 0 : (s_1, a_2),\ R_1 = 1$ | $G \leftarrow 2 + 1 = 3,\ \ C(s_1, a_2) \leftarrow 0 + \frac{5}{2} = \frac{5}{2}$<br>$Q(s_1, a_2) \leftarrow 0 + \frac{\frac{5}{2}}{\frac{5}{2}}(3 - 0) = 3$<br>$w_0 = \frac{0}{0.4} = 0 \Rightarrow W \leftarrow 0$ (stop) |

**Episode 2.**

| | |
|---|---|
| Step $t = 1 : (s_1, a_1),\ R_2 = 3$ | $G \leftarrow 0 + 3 = 3,\ \ C(s_1, a_1) \leftarrow 0 + 1 = 1$<br>$Q(s_1, a_1) \leftarrow 0 + \frac{1}{1}(3 - 0) = 3$<br>$w_1 = \frac{1}{0.6} = \frac{5}{3},\ \ W:\ 1 \mapsto \frac{5}{3}$ |
| Step $t = 0 : (s_1, a_1),\ R_1 = 0$ | $G \leftarrow 3 + 0 = 3,\ \ C(s_1, a_1) \leftarrow 1 + \frac{5}{3} = \frac{8}{3}$<br>$Q(s_1, a_1) \leftarrow 3 + \frac{\frac{5}{3}}{\frac{8}{3}}(3 - 3) = 3$<br>$w_0 = \frac{1}{0.6} = \frac{5}{3},\ \ W:\ \frac{5}{3} \mapsto \frac{25}{9}$ |

**Episode 3.**

| | |
|---|---|
| Step $t = 1 : (s_1, a_1),\ R_2 = 2$ | $G \leftarrow 0 + 2 = 2,\ \ C(s_1, a_1) \leftarrow \frac{8}{3} + 1 = \frac{11}{3}$<br>$Q(s_1, a_1) \leftarrow 3 + \frac{1}{\frac{11}{3}}(2 - 3) = 3 - \frac{3}{11} = \frac{30}{11} \approx 2.7273$<br>$w_1 = \frac{1}{0.6} = \frac{5}{3},\ \ W :\ 1 \mapsto \frac{5}{3}$ |
| Step $t = 0 : (s_2, a_2),\ R_1 = -1$ | $G \leftarrow 2 + (-1) = 1,\ \ C(s_2, a_2) \leftarrow 1 + \frac{5}{3} = \frac{8}{3}$<br>$Q(s_2, a_2) \leftarrow 2 + \frac{\frac{5}{3}}{\frac{8}{3}}(1 - 2) = 2 - \frac{5}{8} = \frac{11}{8} = 1.375$<br>$w_0 = \frac{1}{0.4} = \frac{5}{2},\ \ W :\ \frac{5}{3} \mapsto \frac{25}{6}$ |

## Final accumulators and estimates

$C(s_1, a_1) = 1 + \frac{5}{3} + 1 = \frac{11}{3}, \quad C(s_1, a_2) = \frac{5}{2}, \quad C(s_2, a_2) = 1 + \frac{5}{3} = \frac{8}{3}, \quad C(s_2, a_1) = 0.$

$Q(s_1, a_1) = \frac{30}{11} \approx 2.7273, \quad Q(s_1, a_2) = 3, \quad Q(s_2, a_2) = \frac{11}{8} = 1.375, \quad Q(s_2, a_1) = 0$ (never updated).

**Summary table (final):**

| | $a_1$ | $a_2$ |
|---|---|---|
| $s_1$ | $Q = 30/11,\ C = 11/3$ | $Q = 3,\ C = 5/2$ |
| $s_2$ | $Q = 0,\ C = 0$ | $Q = 11/8,\ C = 8/3$ |

# Off-Policy Monte Carlo Control with Importance Sampling

## Objective

The goal of off-policy Monte Carlo control is to learn an optimal target policy $\pi \approx \pi^*$ while generating experience using a different, exploratory behavior policy $b$. Since data is generated under $b \neq \pi$, importance sampling is required to correct the mismatch between the two policies.

## Algorithm Overview

This method:

- Uses episodic Monte Carlo returns,
- Applies weighted importance sampling,
- Updates action-value estimates $Q(s, a)$,
- Improves a deterministic greedy target policy.

## Key Variables

- $Q(s, a)$: action-value estimate
- $C(s, a)$: cumulative importance weights
- $\pi$: target policy (greedy)
- $b$: behavior policy (soft)

## Weighted Importance Sampling Update

For each step $t$ of an episode (processed backward):

$$G \leftarrow \gamma G + R_{t+1}$$

$$C(S_t, A_t) \leftarrow C(S_t, A_t) + W$$

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)}\big(G - Q(S_t, A_t)\big)$$

The target policy is updated greedily:

$$\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$$

# Early Termination

If the behavior action differs from the target action:

$$A_t \neq \pi(S_t),$$

the update loop is terminated because the importance sampling ratio becomes zero.

# Importance Weight Update

If the loop continues:

$$W \leftarrow \frac{W}{b(A_t \mid S_t)}$$

# Numerical Example

## MDP

- States: $s_1, s_2$
- Actions: $a_1, a_2$
- Discount factor: $\gamma = 1$

## Initial Action-Value Function

| State | Action | $Q(s,a)$ |
|---|---|---|
| $s_1$ | $a_1$ | 0 |
| $s_1$ | $a_2$ | 0 |
| $s_2$ | $a_1$ | 0 |
| $s_2$ | $a_2$ | 0 |

Initial target policy:

$$\pi(s_1) = a_1, \quad \pi(s_2) = a_1$$

Behavior policy:

$$b(a_1 \mid s) = 0.5, \quad b(a_2 \mid s) = 0.5$$

### Observed Episode

| $t$ | State | Action | Reward |
|---|---|---|---|
| 0 | $s_1$ | $a_1$ | 2 |
| 1 | $s_2$ | $a_1$ | 3 |
| 2 | terminal | – | – |

### Backward Updates

At $t = 1$:

$$G = 3, \quad W = 1$$

$$C(s_2, a_1) = 1, \quad Q(s_2, a_1) = 3$$

$$W \leftarrow 2$$

At $t = 0$:

$$G = 5$$

$$C(s_1, a_1) = 2, \quad Q(s_1, a_1) = 5$$

## Final Action-Value Function

| State | Action | $Q(s, a)$ |
|---|---|---|
| $s_1$ | $a_1$ | 5 |
| $s_1$ | $a_2$ | 0 |
| $s_2$ | $a_1$ | 3 |
| $s_2$ | $a_2$ | 0 |

## Key Takeaways

- Off-policy learning separates exploration and optimization.
- Importance sampling corrects policy mismatch.
- Weighted importance sampling reduces variance.
- Greedy updates lead toward optimal policy.

# Worked Example: Off-policy Monte Carlo Control (Weighted Importance Sampling)

## Goal

Demonstrate *one full episode update* using the off-policy MC control algorithm (weighted importance sampling) shown in class:

$$G \leftarrow \gamma G + R_{t+1}, \quad C(S_t, A_t) \leftarrow C(S_t, A_t) + W, \quad Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)}\big(G - Q(S_t, A_t)\big),$$

$$\pi(S_t) \leftarrow \arg\max_a Q(S_t, a), \quad \text{if } A_t \neq \pi(S_t) \text{ then stop back-updating}, \quad W \leftarrow \frac{W}{b(A_t \mid S_t)}.$$

## Example MDP + Policies (given)

**States:** $s_0, s_1$ (terminal state $\texttt{T}$).
**Actions in each state:** $a_0, a_1$.
**Discount:** $\gamma = 0.9$.

### Behavior policy $b$ (used to generate data)

Assume a soft (exploratory) behavior policy:

$$b(a_0 \mid s) = 0.8, \qquad b(a_1 \mid s) = 0.2 \qquad \text{for } s \in \{s_0, s_1\}.$$

### Initial action-values and target policy $\pi$

Initialize $Q(s,a) = 0$ and $C(s,a) = 0$ for all $(s,a)$. Break ties consistently by choosing $a_0$. Hence initially:

$$\pi(s_0) = a_0, \qquad \pi(s_1) = a_0.$$

| **Initial** $Q(s,a)$ | $a_0$ | $a_1$ |
|---|---|---|
| $s_0$ | 0 | 0 |
| $s_1$ | 0 | 0 |

| **Initial** $C(s,a)$ | $a_0$ | $a_1$ |
|---|---|---|
| $s_0$ | 0 | 0 |
| $s_1$ | 0 | 0 |

## Episode generated by following $b$

One episode (length $T = 3$):

$$t = 0: \ (s_0, a_0, r_1 = 1, s_1), \quad t = 1: \ (s_1, a_0, r_2 = 2, s_0), \quad t = 2: \ (s_0, a_0, r_3 = 3, \texttt{T}).$$

## Run the algorithm (backward: $t = 2, 1, 0$)

Initialize $G \leftarrow 0$ and $W \leftarrow 1$.

### Step $t = 2$

$$G \leftarrow 0.9 \cdot 0 + 3 = 3.$$

Update cumulative weight:

$$C(s_0, a_0) \leftarrow 0 + 1 = 1.$$

Update $Q$:

$$Q(s_0, a_0) \leftarrow 0 + \frac{1}{1}(3 - 0) = 3.$$

Greedy improvement (no tie now): $\pi(s_0) = a_0$. Since $A_2 = a_0 = \pi(s_0)$, continue.

$$W \leftarrow \frac{1}{b(a_0 \mid s_0)} = \frac{1}{0.8} = 1.25.$$

### Step $t = 1$

$$G \leftarrow 0.9 \cdot 3 + 2 = 4.7.$$

$$C(s_1, a_0) \leftarrow 0 + 1.25 = 1.25.$$

$$Q(s_1, a_0) \leftarrow 0 + \frac{1.25}{1.25}(4.7 - 0) = 4.7.$$

Thus $\pi(s_1) = a_0$. Since $A_1 = a_0 = \pi(s_1)$, continue.

$$W \leftarrow \frac{1.25}{b(a_0 \mid s_1)} = \frac{1.25}{0.8} = 1.5625.$$

### Step $t = 0$

$$G \leftarrow 0.9 \cdot 4.7 + 1 = 5.23.$$

$$C(s_0, a_0) \leftarrow 1 + 1.5625 = 2.5625.$$

Incremental weighted update:

$$Q(s_0, a_0) \leftarrow 3 + \frac{1.5625}{2.5625}\big(5.23 - 3\big).$$

Compute the factor and result:

$$\frac{1.5625}{2.5625} \approx 0.609756, \quad (5.23 - 3) = 2.23$$

$$Q(s_0, a_0) \approx 3 + 0.609756 \times 2.23 \approx 4.359756 \approx 4.36.$$

Then $\pi(s_0) = a_0$ (still greedy). Since $A_0 = a_0 = \pi(s_0)$, the episode finishes without an early break.

## Final tables after this episode

| **Final** $Q(s, a)$ | $a_0$ | $a_1$ |
|---|---|---|
| $s_0$ | 4.36 | 0 |
| $s_1$ | 4.70 | 0 |

| **Final** $C(s, a)$ | $a_0$ | $a_1$ |
|---|---|---|
| $s_0$ | 2.5625 | 0 |
| $s_1$ | 1.25 | 0 |

**Final greedy target policy** $\pi$**:**

$$\pi(s_0) = a_0, \qquad \pi(s_1) = a_0.$$

## Key observation (why the "early break" exists)

If at some time $t$ the behavior action differs from the current greedy target action, i.e., $A_t \neq \pi(S_t)$, then for a deterministic greedy target policy:

$$\pi(A_t \mid S_t) = 0 \;\Rightarrow\; \rho = 0 \;\Rightarrow\; W \text{ becomes } 0,$$

so earlier steps would contribute nothing meaningful. Hence the algorithm stops back-updating.

# Applications of Off-Policy Monte Carlo Control

## Why Off-Policy Monte Carlo Control?

Off-policy Monte Carlo (MC) control is particularly valuable in real-world settings where the policy being optimized (target policy) cannot be safely or directly deployed during learning. Instead, data is collected using a different, often safer or more exploratory behavior policy. Importance sampling enables learning optimal behavior from such data, making off-policy MC control highly relevant for working professionals.

## Application 1: Learning from Logged User Interaction Data

**Context:** Recommendation systems, search engines, and conversational agents often rely on large volumes of historical interaction logs.

**Why off-policy?** The deployed system (behavior policy) generates data, while engineers want to evaluate or improve a new policy without deploying it live.

**Relevance:** Off-policy MC control allows learning improved action-selection strategies purely from logged trajectories.

**References:**

- Li et al., "Unbiased Offline Evaluation of Contextual-bandit-based News Article Recommendation", WSDM 2011.
- https://arxiv.org/abs/1003.5956

## Application 2: Robotics and Autonomous Systems

**Context:** Robots often operate under conservative controllers for safety.

**Why off-policy?** Exploration-heavy policies are unsafe to execute directly on physical hardware.

**Relevance:** Off-policy MC control enables learning optimal control policies from data generated by safe baseline controllers or simulators.

**References:**

- Sutton et al., "Reinforcement Learning for Robotics", IEEE Robotics & Automation Magazine.

- https://ieeexplore.ieee.org/document/7989602

## Application 3: Healthcare Decision Support

**Context:** Treatment policies are derived from historical clinical practice.

**Why off-policy?** It is unethical or impractical to experiment with arbitrary policies on patients.

**Relevance:** Off-policy MC control allows evaluation and improvement of treatment policies using retrospective patient trajectories.

**References:**

- Gottesman et al., "Guidelines for Reinforcement Learning in Healthcare", Nature Medicine, 2019.
- https://www.nature.com/articles/s41591-019-0512-0

## Application 4: Finance and Algorithmic Trading

**Context:** Trading strategies are tested using historical market data.

**Why off-policy?** A new strategy cannot influence past market behavior.

**Relevance:** Off-policy MC control supports learning and comparing trading policies using logged price-action trajectories.

**References:**

- Moody and Saffell, "Learning to Trade via Direct Reinforcement", IEEE Transactions on Neural Networks.
- https://ieeexplore.ieee.org/document/846882

## Application 5: Conversational AI and Dialogue Management

**Context:** Chatbots operate under fixed dialogue strategies in production.

**Why off-policy?** Experimenting with new dialogue policies live may degrade user experience.

**Relevance:** Off-policy MC control enables learning better dialogue actions from existing conversation logs using importance sampling.

**References:**


- Jaques et al., "Way Off-Policy Batch Deep Reinforcement Learning of Implicit Human Preferences", ICML 2020.
- https://arxiv.org/abs/1907.00456


## Summary

Off-policy Monte Carlo control is especially suitable when:

- Data is abundant but policy deployment is costly or risky,
- Exploration must be separated from optimization,
- Learning must occur from historical or third-party data.

These characteristics make it a natural fit for many industry-scale systems used by working professionals.